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The effect of prenatal environment on brain metabolic function and perception in a lizard

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Abstract

1. The ability to perceive prey is crucial for survival in many animals. However, prey detection is a cognitive process that can be influenced by early life conditions such as prenatal stress or temperature. Mitochondrial function is likely to underly these effects because cognitive abilities are energetically demanding (i.e., dependent on efficient metabolic function) and because metabolic-derived oxidative stress and oxidative damage can impair cognitive function. Yet, whether prenatal stress and thermal environments affect prey detection, and whether mitochondrial function mediates such effects, remains poorly understood.

2. Here, we investigated the links between prenatal environments, mitochondrial function, and prey detection in the delicate skink (*Lampropholis delicata*). We manipulated egg corticosterone (CORT) levels—a key stress hormone in reptiles—and incubation temperature and assessed the ability of lizards to detect chemical and visual prey stimuli. Then, using flow cytometry, we measured metabolic function, oxidative stress, and oxidative damage in the olfactory bulbs and optic tecta, two brain regions involved in sensory processing.

3. While metabolic function and oxidative stress remained robust to early life conditions, CORT and temperature interacted to influence oxidative damage. Particularly, CORT decreased DNA damage in the olfactory bulbs of cold-incubated lizards but had no effect hot-incubated lizards. In both brain regions, lipid peroxidation was lower in CORT-treated lizards incubated at high temperatures compared to CORT-treated cold-incubated lizards, but no other between-treatment differences were detected. Additionally, CORT-treated lizards responded faster when presented with chemical cues but not visual stimuli. We did not find any significant relationship between mitochondrial function and prey detection.

4. Our findings suggest that prenatal conditions can have lasting effects on brain physiology and perception, but mitochondrial function in the olfactory bulbs and optic tecta is not associated with prey detection.

Keywords

Corticosterone, Incubation temperature, Mitochondria, Perception, Prey detection, Reptiles

Introduction

Cognition encompasses the ways in which animals acquire, process, and store information, enabling perception, learning, memory, and decision-making (Shettleworth, 2010). Individual variation in cognitive abilities can translate directly into differences in survival and reproductive success, making cognition a key trait for fitness (Dukas, 2004). However, between-individual variation is common in wild populations due to a combination of genetic and environmental factors (Dukas, 2004; Sakata et al., 2000). Early-life conditions, in particular, play a critical role in shaping brain development and cognitive abilities (Amiel et al., 2017; Crino et al., 2014; Sakata et al., 2000; Zhu et al., 2004). Yet, the impact of early-life environments has primarily been studied in the context of learning and memory (Abayarathna & Webb, 2020; Bebus et al., 2016; Crino et al., 2014; Zhu et al., 2004), neglecting how other important cognitive domains such as stimuli perception might be affected (but see Burger, 1998). Perception is an often overlooked, but fundamental cognitive process, critical for prey detection, avoiding predators, and interacting with conspecifics (Burger, 1998; Desfilis et al., 2003; Hairston Jr et al., 1982; Pyke, 1984; Recio et al., 2023; Shettleworth, 2010). Yet, whether early-life conditions shape perception abilities remains largely unknown.

Early-life conditions can influence cognitive abilities through various mechanisms including alterations in gene expression (Zhou et al., 2020), effects on neurotransmitter production and neuroendocrine pathways (Amani et al., 2021), and impacts on brain structure (Amiel et al., 2017). One of the most fundamental mechanisms through which early-life conditions can influence cognition is by affecting brain mitochondrial function (Du et al., 2009; Picard et al., 2018; Picard & McEwen, 2014; Siegel & Albers, 1994). Cognitive processes are energetically demanding, relying heavily on efficient mitochondrial respiration (Alexandrov & Pletnikov, 2022; Mann et al., 2021; McNay et al., 2000). Likewise, high cognitive abilities are typically associated with increased neuron density and functionality (Amiel et al., 2017; Lefebvre, 2011), both of which are impaired by excessive oxidative stress resulting from mitochondrial respiration (Du et al., 2009; Finkel & Holbrook, 2000; Gong et al., 2011; Hoffmann & Spengler, 2018; Zhu et al., 2004). Therefore, impacts on mitochondrial function during early development may have significant consequences for cognitive abilities such as prey detection.

Mitochondria are highly responsive to the environment, and early-life conditions can have short-term and long-lasting consequences on mitochondrial function (reviewed in Gyllenhammer et al., 2020). Stress-related hormones and temperature are particularly relevant for mitochondrial function due to their strong effects on metabolic processes (Crino, Wild, et al., 2024; Sapolsky et al., 2000; Stier et al., 2022). Vertebrates respond to stressful situations through the activation of physiological processes including the production of glucocorticoid (GC) hormones, which are important metabolic regulators (Sapolsky et al., 2000). Similarly, temperature is an important metabolic regulator in ectotherms (Crino, Wild, et al., 2024; Stier et al., 2022). Prenatal GC exposure has been shown to increase oxidative stress and oxidative damage in offspring (Haussmann et al., 2012; Song et al., 2009; Zhu et al., 2004), while incubation temperature shapes mitochondrial efficiency and oxidative damage in ectotherms (Crino, Wild, et al., 2024; Stier et al., 2022; Treidel et al., 2016). In addition, both abrupt thermal changes and sustained elevated temperatures increase GC levels in ectotherms (Crino, Bonduriansky, et al., 2024), with potential interactive effects between prenatal temperature and stress hormones. As such, both prenatal GCs and thermal environment are predicted to have independent and interactive effects on cognitive abilities, such as prey detection, through alterations in mitochondrial metabolism, oxidative stress, and oxidative damage. Yet, the links between prenatal environment, mitochondrial function, and prey detection remain largely unexplored, particularly in reptiles. Here, we investigated the effects of prenatal temperature and corticosterone (CORT)—the main GC in reptiles—on mitochondrial function and prey detection in an Australian lizard, the delicate skink (*Lampropholis delicata*). We manipulated egg CORT levels and incubation temperature in a fully factorial design, measured prey detection ability and then assessed mitochondrial function in brain regions related to the processing of chemical and visual cues. Specifically, we analysed mitochondrial density and metabolic capacity as a proxy for understanding the energetic efficiency of the cell, and reactive oxygen species (ROS) as a proxy for oxidative stress. In addition, we also analysed DNA damage and lipid peroxidation, which are typical damage caused by oxidative stress (hereafter also oxidative damage). We predicted that elevated temperatures and high CORT levels would have sustained effects on mitochondria in the brain, including decreased energetic efficiency and increased oxidative stress and damage (Costantini et al., 2011; Crino, Wild, et al., 2024; Haussmann et al., 2012; Song et al., 2009; Stier et al., 2022; Treidel et al., 2016; Zhu et al., 2004), ultimately impairing prey detection ability (Alexandrov & Pletnikov, 2022; Hara et al., 2014; Picard et al., 2018; Zhu et al., 2004).

Methods

Animal husbandry

Breeding colony – The lizards included in this study came from a breeding colony established in 2019. The colony consisted of 270 adults of *L. delicata* housed in plastic containers (41.5 L x 30.5 W x 21 H cm) with six lizards (two males and four females) per enclosure. Enclosures were provided with non-stick matting, shelter, and several small water dishes. Water was given daily, and lizards were fed ~40 mid-size living crickets (*Acheta domestica*) per enclosure three days a week. Crickets were dusted with calcium weekly and multivitamin and calcium biweekly. Room temperatures were set to 22-24 ºC, but enclosures were also provided with a heat chord and a heat lamp following a 12 h light:12 h dark cycle that maintained on side of the enclosures at ~34 ºC.

Egg collection and incubation – Between mid-October 2022 and the end of February 2023, we provide females with a place to lay the eggs by placing a small box (12.5 L x 8.3 W x 5 H cm) with moist vermiculite in the communal enclosures. We checked the boxes three days a week for eggs. After collection, eggs were treated with either a CORT or a control solution (see CORT and temperature manipulation below) and placed in individual cups (80 mL) with moist vermiculite (12 parts water to 4 parts vermiculite). Cups were covered with cling wrap to retain moisture and left at one of two different incubation temperatures (see CORT and temperature manipulation below) until hatching.

Hatchlings – Incubators were checked three times a week for hatchlings. After hatching, lizards were placed in individual enclosures (18.7 L x 13.2 W x 6.3 H cm) provided with nonstick matting and a small water dish until the beginning of the experiment. During this period, lizards were given water daily and received 3-6 small living crickets three times a week. Remaining care followed similar protocols for adults (see above).

CORT and Temperature manipulation

To test the interactive effects of CORT and incubation temperature, we manipulated CORT concentrations in eggs and incubated them under one of two temperature regimes (see Fig. 1 A). We used a partial split clutch design where eggs from a given clutch were distributed equally across the four treatments when clutch sizes were larger than four and randomly across treatments when less than four. Eggs were topically treated with either: a) 5 µL of CORT solution or b) 5 µL of 100% ethanol (Control treatment). The CORT solution was made by dissolving crystalline corticosterone (Sigma, Cat. No. C2505) in 100% ethanol. The concentration of the CORT treatment (10 pg/mg, CORT relative to average egg mass) was based on a previous study where CORT treatment increased mean yolk CORT levels by approximately 2 standard deviations above the mean natural concentration (Crino, Wild, et al., 2024). Eggs were then incubated at either cold (23 ± 3 ºC) or hot (28 ± 3 ºC) (see Fig. 1). These temperatures are within the natural limits in *L. delicata* (Cheetham et al., 2011).

Prey discrimination tests

Two weeks before we started the tests (see below), lizards were moved to the experimental arena (see Fig. 1) for acclimatisation. The arenas were individual plastic containers (41 L x 29.7 W x 22 H cm) with a shelter (9 L x 6 W x 1.5 H cm) on one of the extremes and a water dish in the middle of the arena. Arenas were placed in two rooms on six racks, each with its own CCTV system (device model DVR-HP210475) to record behavioural trials (see details below). The number of lizards per treatment in each rack was counterbalanced to control for any effect of the room or the position of the lizard on the rack. During acclimatisation, lizards were fed with only one cricket per day dusted with calcium and multivitamin, and water was supplied *ad libitum*. We provided a temperature gradient by means of a heat cord and heat lamps in a 12 h light: 12 h dark cycle. The room temperature was set to between 22-24 Celsius. After the tests, animals were euthanised and metabolic function was analysed in various brain regions (see *Brain mitochondrial function* below).

To measure prey detection, we presented lizards with chemical and visual stimuli from familiar and unfamiliar prey, then recording and analysing their behaviour towards each stimulus (Fig. 1 C). We used living crickets (*A. domestica*) as the familiar prey and living mealworm larvae (*Tenebrio molitor*) as the unfamiliar prey. We tested the effects of familiarity with the pretty items because previous experience may influence stimuli perception through habituation or sensitisation (Burger, 1990, 1991; Desfilis et al., 2003).

Each prey item was presented inside a transparent plastic vessel containing a white, two-chambered device (see Fig. 1 B) made of polylactic acid (PLA). In chemical trials, the prey item was placed in the closed chamber at the back of the device, making it invisible to the lizard, while in visual trials, the prey item was placed in the front chamber. Holes in both the device and the front sides of the transparent vessel (see Fig. 1 B, C) allowed chemical cues to be released; however, these holes were sealed with silicone in the visual trials. To increase the availability of chemical cues, we glued a piece of filter paper (left for at least 8 hours in one of the prey's enclosures: *A. domestica* or *T. molitor*) to the device during chemical trials. In visual trials, the filter paper was placed in an empty box for the same duration under identical conditions. In both chemical and visual trials, the prey remained inside the vessel to control for potential acoustic cues. The order of stimulus presentation was counterbalanced across treatments.

Each trial began by placing the experimental device at the side of the arena opposite to the shelter (see arena in Fig. 1 A) and removing the water cup and the shelter. We recorded the lizard's behaviour for approximately one hour. We assessed lizards' ability to detect each stimulus ('Detection latency') by recording the time from when the lizard resumed normal activity (i.e., walking for at least 5 consecutive seconds; T_0 in Fig. 1 D) until the first interaction with the stimulus (T_D in Fig. 1 D). Lizards were considered to have interacted with the object when the lizard touched the front of the vessel or the filter paper with its snout for more than five consecutive seconds.

To control for potential differences in hunger levels, all lizards were fasted for two days before the experiment, a period considered harmless for this species (Young et al., 2022). After each trial, the lizards were also given a cricket to assess their motivation to forage. The cricket was left in the enclosure for one hour, and we recorded whether the lizard ate it (recorded as 1) or not (recorded as 0). In 23 videos, the camera stopped recording before the end of the motivation test (T_f in Fig. 1 D), so motivation was recorded as NA. We used their performance in the motivation test as a covariate in the analyses (see below).

All trials were conducted between 1100 and 1300 h, when the lizards were most active. To control for potential effects of neophobia, we simulated test conditions for two days prior to the experiment by removing the shelter and water cup and exposing the lizards to the vessel without a stimulus. This simulation lasted for 1 hour at the same time of day as the tests, but no behavior was recorded.

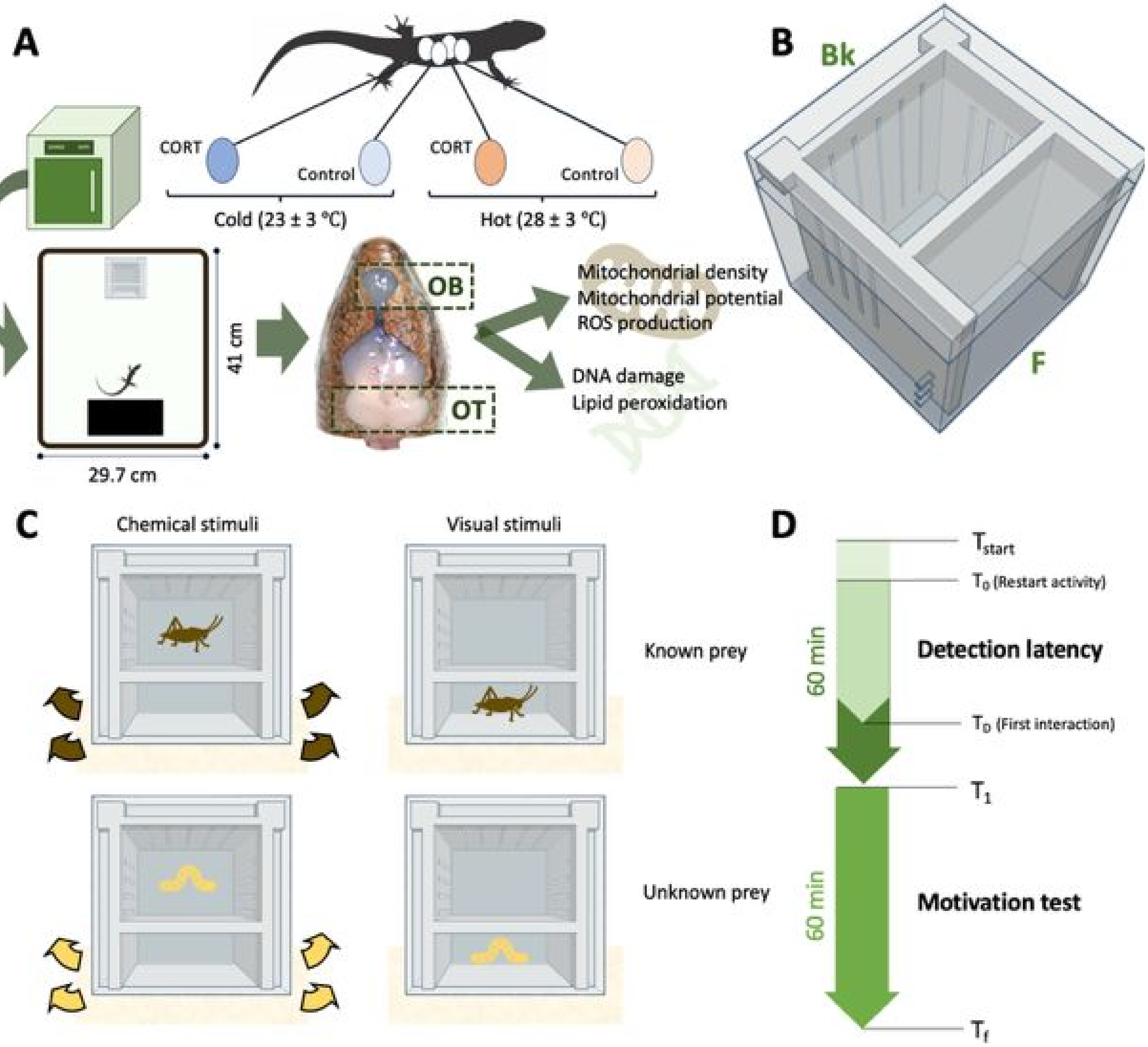

Fig 1— Scheme of our experimental design. In panel A, we show the different stages of our experiment and the main manipulations. In panel B, we show the experimental device used to present the stimuli in the behavioural tests. Here, F indicates the front of the device, and Bk the back. In panel C, we show the experimental setup for the prey discrimination tests. Arrows here indicate the presence of chemical cues. In panel D, we show the relevant times from our behavioural tests.

Brain mitochondrial function

Brain dissection and homogenisation: Two months after the completion of the tests, we euthanised lizards using an injectable anaesthetic followed by decapitation. We injected intraperitoneally 10 mg/kg of a 10 mg/mL alfaxan solution and then, after several minutes, we evaluated the lizard’s righting response and pinching reflex in one of the front limbs. Lizards without responses were decapitated with surgical scissors. The procedure lasted between two to five minutes for each lizard. This protocol was approved by the [redacted].

After decapitation, the head was opened and the brain was dissected. We extracted two main regions of the brain, the olfactory bulbs and the optic tecta, as they are associated with chemical and visual perception in lizards (Wyneken, 2007). Both regions were transferred immediately to 1.5mL centrifuge tubes containing 100μL of 1X phosphate buffered saline (PBS) and homogenised.

For the olfactory bulbs, the tissue was mechanically homogenised by placing the tissue in the well of a 100 μm mesh filter (pluriStrainer) affixed atop a 1.5 mL centrifuge tube, then mashed with the rubber end of an insulin syringe stopper. The resulting olfactory bulb homogenate was then rinsed through a filter with 1 mL of cold 1X PBS. Optic tecta were mechanically homogenised the same way, but were first enzymatically digested by incubating the tissue in 100μL of 125 U/mL collagenase (type II) for 30 min at 32°C. Homogenates were split among two aliquots: one was used within two hours of collection to measure mitochondrial density, mitochondrial membrane potential—a metric of metabolic capacity (Martínez-Reyes et al., 2016)—and ROS, while the other one was cryopreserved for later measurements of DNA damage and lipid peroxidation. Cryopreservation was made by suspending the homogenates in 1 mL solution of 1% Neutral-Buffered Formalin, 1X Tris-EDTA, and 10% DMSO, then stored at -20 °C until NA damage and lipid peroxidation assays.

Mitochondrial density, metabolic capacity, and ROS dyes: Fresh homogenates were stained with 5 μL of a fluorescent probe mix containing equal parts 5 μM MitoTracker Deep Red FM, 2.5 μM MitoTracker Orange CMTMRos, and 50 μM MitoSOX Red to measure mitochondrial density, metabolic capacity, and superoxide (ROS) production, respectively. We also added 10 μg/mL Hoechst 33342 Nuclear Viability Dye to each sample, which we used to distinguish viable cells from debris. Within two hours after dissection, these samples were analysed by flow cytometry (see below).

DNA damage and lipid peroxidation dyes: Assays of DNA damage and lipid peroxidation from cryopreserved samples were performed 61 weeks after sample collection. On the day of the assays, we rapidly thawed frozen samples, centrifuged each sample at 1000 RCF for 10 minutes to pellet cells, then removed the supernatant, and resuspended samples in 1 mL 1X Tris-EDTA.

We then centriduged the samples again and removed the Tris-EDTA, resuspending the samples in 200 μL of 1X PBS containing 10 μg/mL Hoechst 33342 Nuclear Viability Dye (for cell viability) and 100 μM BODIPY 665/676 Lipid Peroxidation Sensor (to measure lipid peroxidation). Following staining, we permeabilised the cell membranes incubating the samples in 200 μL 1X PBS containing 20 μM digitoninad then stained the samples with 20 μL of 70 μM 8-OHdG Polyclonal Antibody to measure DNA damage. We left the homogenate overnight (~12 hours) at 4 ºC. The following day we counterstained the cells with 20 μL of 100 μg/mL H+G Goat Anti-Rabbit Conjugate Antibody with Alexa-Fluor 488 and analyse the samples in the flow cytometer.

Flow Cytometry: Flow cytometry assays were performed using a Becton Dickson LSRFortessa X-20 flow cytometer with the default wavelength filters on detectors. The detectors and voltage settings used in data acquisition for each were determined during pilot trials and kept consistent throughout the experiment. Data was imported into FlowJo (v. 10.1) for processing. For further details on the homogenisation, staining, or flow cytometry assays, see *Methods: flow cytometry in Supplementary Material*. We validated that our homogenates contained neurons in a pilot study using dyes specifically targeting neuronal nuclei (see *Brain validation in Supplementary Material*). This pilot study also ensured that our gating strategy identified these neurons using flow cytometry.

We recorded five physiological variables from our flow cytometry assay: mitochondrial density, metabolic capacity, ROS production, DNA damage, and lipid peroxidation. These variables reflect average per-cell values for each sample, as fluorescence intensity was normalized by the number of gated events (cells) recorded. However, to assess how early environmental conditions may influence mitochondrial function at the organelle level, we created two additional response variables by dividing metabolic capacity and ROS production by mitochondrial density. This procedure assumes that mitochondrial density serves as a proxy for the number of mitochondria per cell, and the derived variables provide an estimate of the average activity per mitochondrion, therefore allowing us to assess mitochondrial-specific responses to early-life conditions. However, we report also the results for metabolic capacity and ROS production alone to understand the effects of early-life condition on metabolic function at the cell level.

Replication Statement

Scale of inference	Scale at which the factor of interest is applied	Number of replicates at the appropriate scale
Individual	Egg (CORT manipulation)	20 per treatment combination
Individual	Incubator (temperature manipulation)	4 incubators (two per temperature treatment)

Statistical analyses

To investigate the effects of prenatal conditions on mitochondrial function and prey detection, we performed the analyses for each brain region/stimulus and each variable separately, running a total of 16 different models. For every model, we first fit a set of preliminary models where we included the hormone (CORT versus Control), temperature (Cold versus Hot), and their interaction. These models also included the sex and age of the lizards at the time where the trials started (for detection latency) or when the lizards were euthanised (for all mitochondrial-related variables). For detection latency, we also included previous experience with the prey (familiar versus unknown), their performance on the motivation test [if they ate the cricket (1), or not (0)], and the interaction between CORT and motivation as fixed effects. We included the interaction between CORT and motivation because CORT can impact appetite (Conde-Sieira et al., 2018). All models included clutch identity as a random factor to account for clutch effects. For models of detection latency we also included a random effect of lizard identity because of our repeated measures design. The structure and results of these models are provided on *Tables S1-S22 in Supplementary Material*. After the preliminary models, we simplified models by excluding some of the factors that were not significant. However, we always included the main interest factors: CORT, temperature and their interaction. Random factors remain the same for the final models. All the response variables were mean centered and standardized by dividing by two times the standard deviation (Gelman, 2008). Before standardization, mitochondrial density, DNA damage, lipid peroxidation, and detection latency were log-transformed.

All models were fit using using the package *brm* in Stan (Stan Development Team, 2024) in R (version 4.4. 0) (R Core Team, 2021). We chose a Bayesian framework because it provides more reliable parameter estimates and direct probability statements about effects, and because it allows simultaneous estimation of all SEM paths with full uncertainty propagation. The error structure was modelled assuming a Gaussian distribution for all variables. We ran four parallel MCMC chains of 8000 iterations for each model, with a warmup period of 2000 iterations. To test for differences between treatments, we made contrasts between treatments using the posterior distribution of relevant parameters. Significance was assessed by using the posterior distributions of parameter estimates and contrasts to test whether were they different from zero (Endo et al., 2019). We considered an effect statistically significant if pMCMC < 0.05.

To further explore the relationships between mitochondrial physiology and detection latency, we used a Structural Equation Modelling (SEM) approach. We fit a multivariate brm in stan (Stan Development Team, 2024) in R (version 4.4.0) (R Core Team, 2021) for each brain region/stimulus separately. We included in the model all the variables of interest and their interactions structured following a specific set of hypotheses (see Fig. 5 and Fig. 6). Because experience with prey did not affect lizards' behaviour (see *Tables S21, S22 in Supplementary Material*), and to reduce the complexity of the models, we averaged lizards detection latency across both types of prey and excluded lizard identity from the random factors. Clutch identity was included as a random factor for all variables. For our SEM model, the error structure was modelled assuming a Gaussian distribution and we estimated residual correlations among all variables.

We obtained direct, indirect, and total effects from posterior distributions of parameters estimated in the multivariate models. Direct effects represent the posterior estimates of a predictor's effect on the response variable, while indirect effects were computed as the product of the direct effects along the path (Kline, 2005). Total effects are the sum of direct and indirect effects.

Results

Our final sample size for mitochondrial assays was 80 lizards from a total of 50 clutches (n = 20 per treatment). Each lizard was subjected to 4 tests for a total of n = 320 behavioural observations (n = 2 with missing data).

Do prenatal CORT and temperature affect mitochondrial function in the brain?

Models explained between 37.8 to 50.4 % of the variation in mitochondrial density and metabolic capacity and 39.9 to 59 % for ROS production and oxidative stress (see Table S1).

Olfactory bulbs: DNA damage and lipid peroxidation increased with age (Table S3), but neither prenatal CORT or temperature independently affected any mitochondrial variable analysed (see Fig. 2, Fig. 3, and Table S2). Lipid peroxidation, however, varied depending on the prenatal CORT treatment and incubation temperature: although the interaction was weak ([($\beta_{Control\text{-}Hot}$ -

$\beta_{CORT\text{-}Hot}$) - ($\beta_{Control\text{-}Cold}$ - $\beta_{CORT\text{-}Cold}$)]: mean = 0.125, pMCMC =0.344), targeted contrasts showed CORT-treated lizards incubated at high temperatures had lower peroxidation than CORT-Cold lizards ($\beta_{CORT\text{-}Hot}$ - $\beta_{CORT\text{-}Cold}$: mean = -0.242, pMCMC < 0.05; see Fig. 3 D).

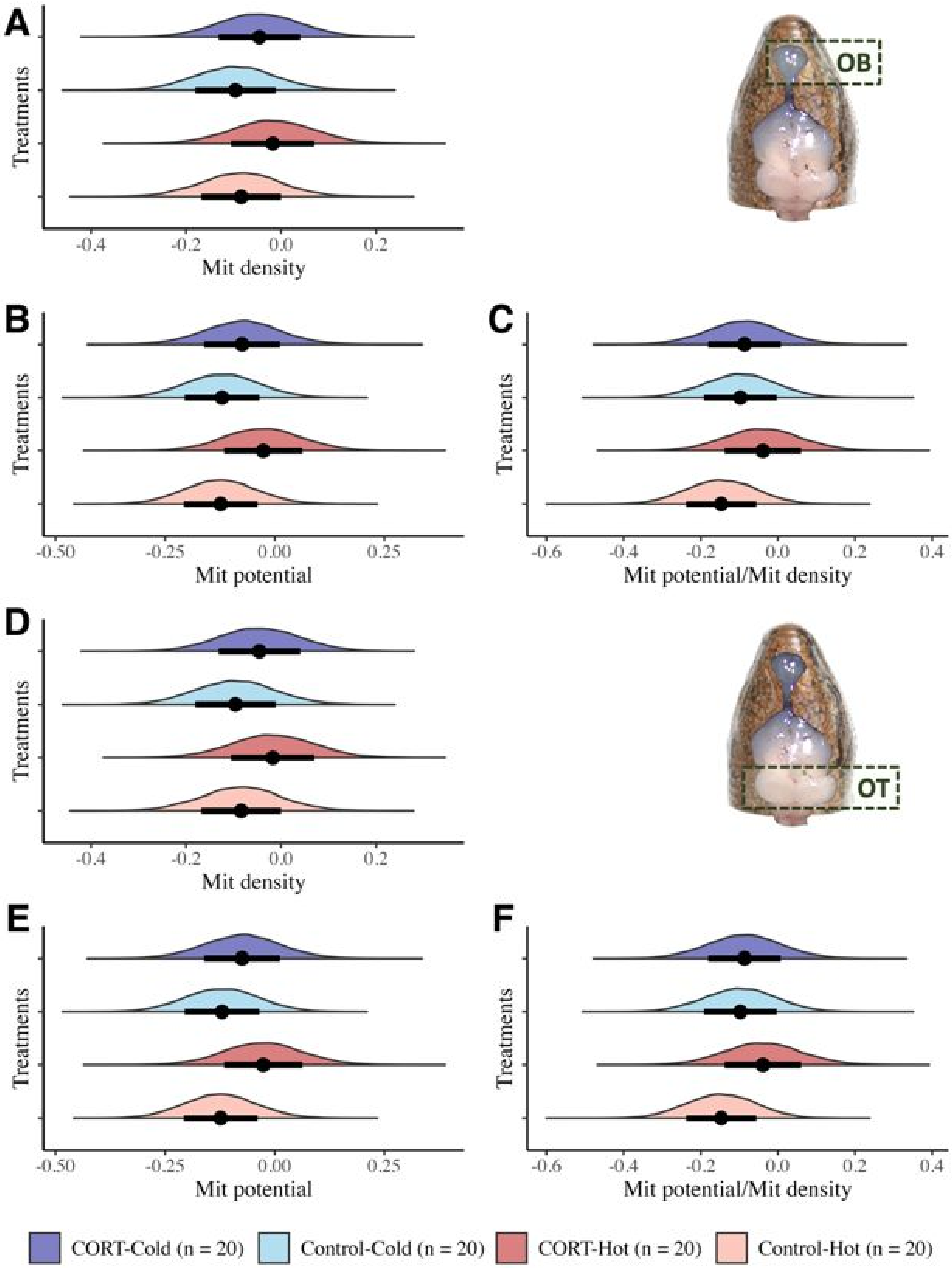
A
Treatments
-0.4
-0.2
0.0
0.2
Mit density
OB
B
Treatments
-0.50
-0.25
0.00
0.25
Mit potential
C
Treatments
-0.6
-0.4
-0.2
0.0
0.2
0.4
Mit potential/Mit density
D
Treatments
-0.4
-0.2
0.0
0.2
Mit density
OT
E
Treatments
-0.50
-0.25
0.00
0.25
Mit potential
F
Treatments
-0.6
-0.4
-0.2
0.0
0.2
0.4
Mit potential/Mit density
CORT-Cold (n = 20)
Control-Cold (n = 20)
CORT-Hot (n = 20)
Control-Hot (n = 20)

Fig 2— Estimates of mitochondrial density (A, D) and metabolic capacity (B, E), and metabolic capacity per mitochondrion (C, F) in the olfactory bulbs (A-C) and optic tecta (D-F) of L. delicata as a function of the different prenatal conditions. Black dots indicate the posterior mean, and the bars represent the SD of the estimates. The y-axis represents the posterior estimates of the variable of interest, and the x-axis represents the different prenatal conditions. Lines with asterisks represent significant differences between groups based on pMCMC values (pMCMC < 0.05), no lines indicate no significant differences.

Optic tecta: Neither prentatal CORT, temperature, or their interaction influenced mitochondrial density, metabolic capacity, or ROS production, and prenatal CORT and temperature did not have independent effects on oxidative damage (see Fig. 2, Fig. 3, and Table S2). However, the interaction of CORT and temperature affected DNA damage ([($\beta_{Control-Hot}$ - $\beta_{CORT-Hot}$) - ($\beta_{Control-Cold}$ - $\beta_{CORT-Cold}$)]: mean = -0.211, pMCMC < 0.05), and lipid peroxidation ([($\beta_{Control-Hot}$ - $\beta_{CORT-Hot}$) - ($\beta_{Control-Cold}$ - $\beta_{CORT-Cold}$)]: mean = 0.217, pMCMC < 0.05). While CORT elevations had no effect on hot-incubated lizards ($\beta_{Control-Hot}$ - $\beta_{CORT-Hot}$: mean = 0.026, pMCMC =0.612), it decreased DNA damage in cold-incubated animals ($\beta_{Control-Cold}$ - $\beta_{CORT-Cold}$: mean = 0.236, pMCMC < 0.05) (see Fig. 3 C, E). The effects of CORT on lipid proxidation depended on incubation temperature: CORT tended to reduce lipid peroxidation in hot-incubated lizards ($\beta_{Control-Hot}$ - $\beta_{CORT-Hot}$: mean = 0.102, pMCMC =0.188) but increase in it at low incubation temperatures ($\beta_{Control-Cold}$ - $\beta_{CORT-Cold}$: mean = -0.115, pMCMC =0.121), altghout these patterns were not statistically supported. However, CORT-Hot lizards had lower lipid peroxidation values than CORT-Cold lizards ($\beta_{CORT-Hot}$ - $\beta_{CORT-Cold}$: mean = -0.193, pMCMC < 0.05; see Fig. 3 C). In addition, we found that females had higher DNA damage levels than males, and lipid peroxidation increased with age (see Table S4).

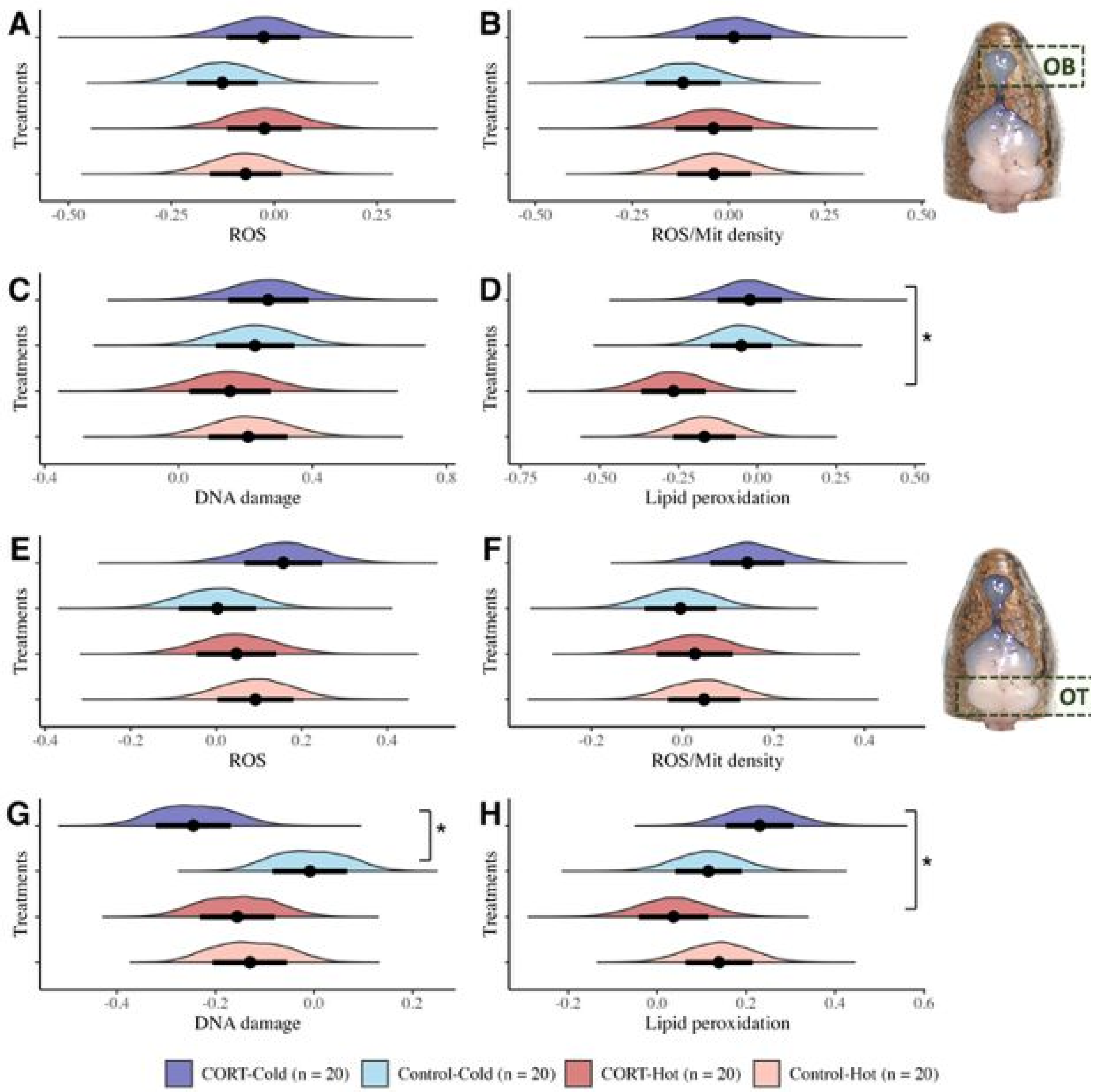

Fig 3— Estimates of ROS (A, E), ROS per mitochondrion (B, F), DNA damage (C, G), and lipid peroxidation (D, H) in the olfactory bulbs (A-D) and optic tecta (E-H) of L. delicata hatchlings as a function of the different prenatal conditions. Black dots indicate the posterior mean, and the bars represent the SD of the estimates. The y-axis represents the posterior estimates of the variable of interest, and the x-axis represents the different prenatal conditions. Lines with asterisks represent significant differences between groups based on pMCMC values (pMCMC < 0.05), no lines indicate no significant differences.

Do prenatal CORT and incubation temperature affect prey detection?

Models explained between 26 to 29.3 % of the variation in prey detection (see Table S1). *Chemical stimulus*: Lizards detected chemical stimulus faster when exposed to prenatal CORT ($\beta_{Control}$ - β_{CORT}: mean = 0.237, pMCMC $<$ 0.05), but there was no effect of temperature or the interaction between CORT and temperature (Fig. 4 A and Table S2).

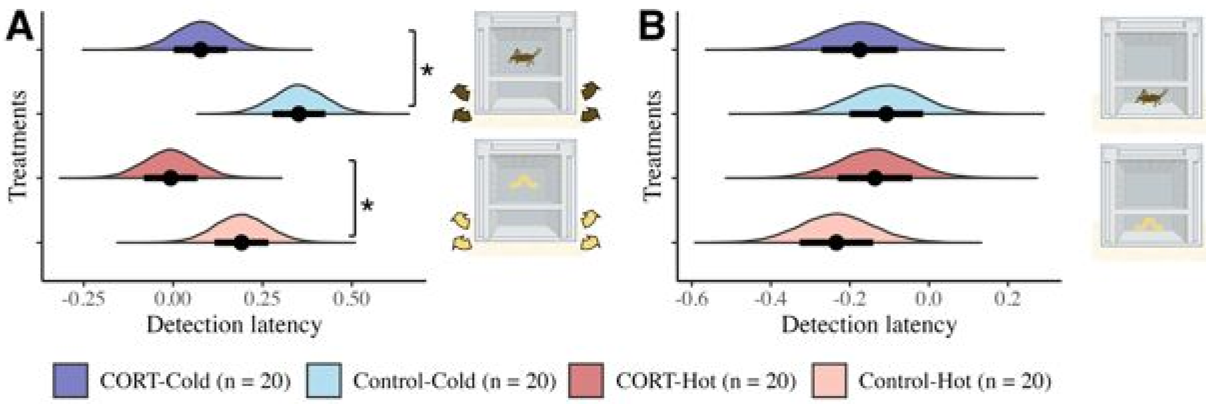

Fig 4— Estimates of detection latency of chemical (A) and visual (B) stimulus by L. delicata hatchlings as a function of the different prenatal conditions. Black dots indicate the posterior mean, and the bars represent the SD of the estimates. The y-axis represents the posterior estimates of the variable of interest, and the x-axis represents the different prenatal conditions. Lines with asterisks represent significant differences between groups based on pMCMC values (pMCMC $<$ 0.05), no lines indicate no significant differences.

Visual stimulus: There were no significant effects of CORT, temperature, or their interaction on the detection latency of visual stimuli (Fig. 4 B and Table S2).

Does mitochondrial function compromise prey detection?

Chemical stimulus (Olfactory bulbs): We found a positive effect of mitochondrial density on ROS production (mean β~Mit density~ $_{\text{- ROS}}$ = 0.864, pMCMC =0.080). However, there were no significant relationships between other mitochondrial variables or between mitochondrial function and prey detection (see Fig. 5 and Table S5).

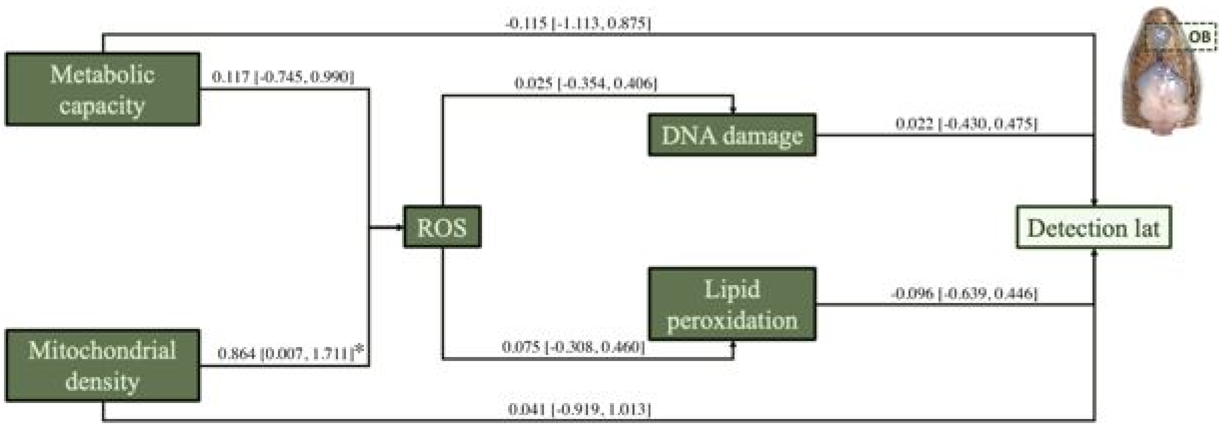

Fig 5— Structural equation model for the olfactory bulbs (OB) and chemical prey detection. Numbers on arrows represent estimated direct effects (mean and 95% credible interval). Asterisks indicate effects whose 95% credible interval does not overlap zero.

Visual stimulus (Optic tecta): We found no significant relationships within the mitochondrial variables analysed or between mitochondrial function and cognitive performance (see Fig. 6 and Table S6).

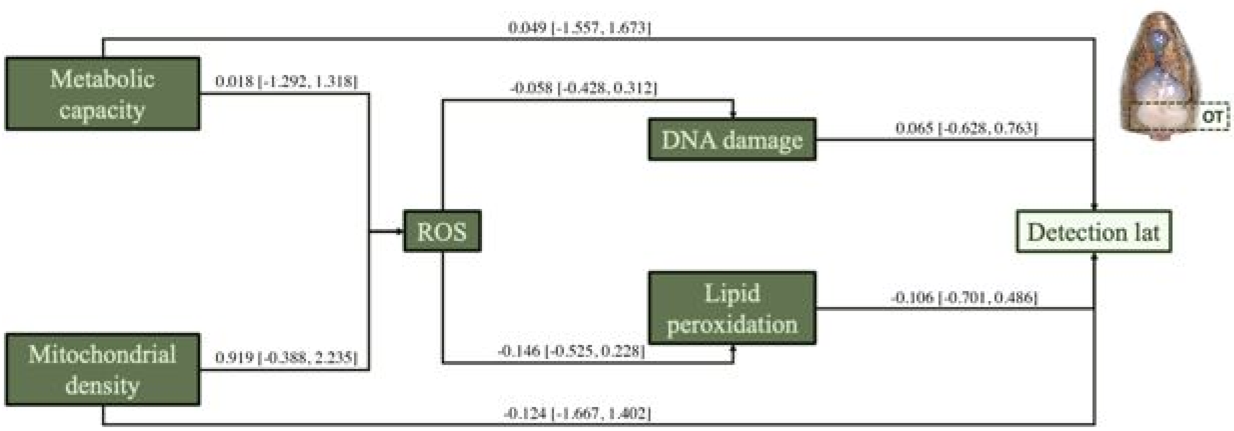

Fig 6— Structural equation model for the optic tecta (OT) and visual prey detection. Numbers on arrows represent estimated direct effects (mean and 95% credible interval). Asterisks indicate effects whose 95% credible interval does not overlap zero.

Discussion

The prenatal environment can impact the development of cognitive abilities through mitochondrial function. However, the links between prenatal environment, prey detection, and mitochondrial function remain largely unexplored. Here, we examined the impact of early-life conditions on perception and its underlying physiological mechanisms. In particular, we analysed how prenatal CORT and temperature influenced the ability of lizards to detect visual and chemical stimuli from prey, and how these developmental effects were associated with alterations in metabolic function (mitochondrial density and metabolic capacity), oxidative stress (ROS), and oxidative damage (DNA damage and lipid peroxidation). Overall, metabolic function and oxidative stress appeared largely resilient to prenatal conditions. However, prenatal conditions produced region- and context-dependent effects on oxidative damage and prey detection. We observed interactive effects of prenatal CORT and incubation temperature on lipid peroxidation in the two brain regions examined, while the interaction affected DNA damage only in the optic tecta. Behaviourally, prenatal CORT exposure enhanced detection of chemical but not visual prey stimuli. Yet, we found no evidence that mitochondrial function underpinned prey detection in *L. delicata*, suggesting a surprising decoupling between brain physiology and perception in this species.

Metabolic function and oxidative stress are robust to early life conditions

Contrary to our predictions, we found no significant effects of early environmental conditions on mitochondrial density, metabolic capacity (mitochondrial membrane potential), or ROS production in the optic tecta or the olfactory bulbs. These results were consistent when metabolic capacity and ROS production were standardised by mitochondrial density, showing no effects of prenatal conditions neither at the cellular nor mitochondrial level. Our findings contrast to previous results showing that mitochondrial efficiency in liver tissue was higher in *L. delicata* incubated at cooler incubation temperatures (Crino, Wild, et al., 2024). The differences between the study by Crino, Wild, et al. (2024) and ours could arise for multiple reasons. For instance, we conducted our mitochondrial analyses using flow cytometry instead of a respirometry, and considered the mitochondrial membrane potential as a proxy of cellular metabolic capacity and the mitochondrial membrane potential/mitochondrial density as a proxy for mitochondrial metabolic capacity. These methodological nuances capture different aspects of mitochondrial function and could explain the differences in our study compared to Crino, Wild, et al. (2024). However, those differences could also reflect tissue-specific responses to prenatal conditions, as the brain can buffer the effects of adverse prenatal environment (Chaudhari et al., 2022). Nonetheless, there is also evidence that the effects of prenatal environment on brain physiology can be region-specific (Coomber et al., 1997). Consistent with this, other studies show that the effects of prenatal environment on brain physiology can be region-specific (Coomber et al., 1997). Hence, other tissues or brain nuclei in *L. delicata* may be more sensitive to prenatal CORT and temperature.

Alternatively, prenatal conditions may have affected mitochondrial density, metabolic capacity, and ROS production during incubation, but these effects were not sustained once the environmental alteration was removed. Mitochondria are highly responsive to environmental changes and energetic demands (Gyllenhammer et al., 2020), and given that all lizards experienced identical post-hatching conditions, any differences in these aspects of mitochondrial function during incubation may have dissipated by the time of measurement at 238-356 days post-hatching (mean = 273 days). Nonetheless, even temporary alterations during earlier stages would still be expected to affect oxidative damage, which accumulates over time (Rice et al., 2002; Terman & Brunk, 2006).

The effects of prenatal CORT on oxidative damage are region- and temperature- dependent

We found that DNA damage and lipid peroxidation were affected in region-specific ways. Early conditions did not influence DNA damage in the olfactory bulbs, but they did alter lipid peroxidation in the olfactory bulbs and the optic tecta. In the optic tecta, prenatal CORT reduced DNA damage in cold-incubated lizards but had no effect in hot-incubated lizards. In both regions, CORT also interacted with incubation temperature to influence lipid peroxidation. Lizards treated with prenatal CORT and incubated at cold temperatures had higher levels of lipid peroxidation compared to the lizards from the CORT-Hot group. These contrasting patterns suggest that prenatal CORT and temperature may shape oxidative damage through molecule-specific protective pathways, potentially by upregulating different antioxidant systems. For example, prenatal stress in rats increases Cu/Zn superoxide dismutase (SOD) and glutathione peroxidase (GPx) activity in the brain while catalase (CAT) was unaffected (McIntosh et al., 1998). Similarly, in the soft-shelled turtle (*Pelodiscus sinensis*), CAT expression elevates with increasing temperatures as turtles emerge from hibernation, whereas GPx activity remains stable (Tang et al., 2021). Such differential regulation of antioxidant systems could explain why CORT reduces DNA damage under cold incubation but reduces lipid peroxidation under hot incubation. However, other mechanisms could explain the CORT-temperature interaction on oxidative damage. CORT mobilises energy resources to facilitate physiological adjustments to stress (Picard et al., 2018; Picard & McEwen, 2014; Sapolsky et al., 2000), and moderate increases in metabolic rate can act as a signal to upregulate antioxidant defences and DNA repair (Kawamura et al., 1991; but see McIntosh et al., 1998). At cold incubation temperatures, CORT may have moderately increased metabolic rate, increasing antioxidant production and, consequently, reducing oxidative damage. However, under higher incubation temperatures, metabolic rate may have risen to exceed the threshold at which antioxidant production could counteract ROS, leaving tissues more vulnerable to oxidative damage. This temperature-dependent ceiling on protective capacity could explain the region-specific patterns we observed, though it does not fully account for reduced lipid peroxidation in lizards exposed to prenatal CORT and hot incubation temperatures. Additional factors, such as temperature-dependent differences in membrane lipid composition (Zeis et al., 2019), may further modulate molecule-specific sensitivity to oxidative stress. For example, lipid peroxidation levels have been shown to increase in cold-acclimated *Daphnia magna* likely because of higher concentration of polyunsaturated fatty acids, which are more susceptible to oxidative stress (Zeis et al., 2019). Future studies should examine the effects of prenatal CORT and temperature on metabolic rate, antioxidant regulation, and molecule structure and composition in different brain nuclei of reptiles.

Prenatal CORT exposure enhances chemical but not visual prey detection

Lizards exposed to prenatal CORT were faster at detecting chemical, but not visual, prey cues compared to control lizards. This effect was not related to alterations in mitochondrial function in the olfactory bulbs. Instead, prenatal CORT could influence other mechanisms involved in chemoreception. For instance, in rats (*R. norvegicus*), dexamethsone (a synthetic GC) enhances the ability to detect chemical stimuli by upregulating gene expression in the olfactory mucosa (Meunier et al., 2020). If similar GC-dependent transcriptional changes occur in lizards, they could alter chemosensory sensitivity independent of mitochondrial function.

Alternatively, the observed differences in chemical perception may be driven by treatment effects on motivation. Prenatal CORT can increase hunger through increased metabolic rates and resource mobilisation (Cossin-Sevrin et al., 2022; Spencer & Verhulst, 2008; but see Crino et al., 2014), potentially elevating motivation in CORT-treated lizards. Although we fasted lizards for two days to standardised hunger levels and the interaction between CORT and motivation was not significant in our models (Table S21), higher motivation may have affected lizards' behaviour beyond what we could control for. For instance, higher motivation may have induced a shift in foraging strategy towards more active searching behaviour. Active foraging lizards rely heavily on chemosensory exploration (Verwaijen & Van Damme, 2007), suggesting that motivationally-induced shifts in foraging mode could enhance chemical but not visual prey detection. Further studies need to be conducted to understand how CORT can change foraging strategies in lizards and how this can affect their perception abilities and decision-making.

Mitochondrial function and oxidative stress do not affect detection abilities

Mitochondria provide energy necessary for cognitive process and influences neuron death and senescence through the production of ROS (a byproduct of cellular respiration; Alexandrov & Pletnikov, 2022; Du et al., 2009; Mann et al., 2021; McNay et al., 2000; Picard et al., 2018; Picard & McEwen, 2014; Siegel & Albers, 1994). Hence, a strong relationship between mitochondrial function and cognitive abilities—such as prey detection—was expected. Indeed, links between brain metabolic function and visual acuity or olfactory capacity have been reported in other systems (reviewed in Wong-Riley, 2010). For instance, energetic deficiencies caused reduced cytochrome c oxidase (COX) activity in dopaminergic neurons of the olfactory bulb of mice (*Mus musculus*) are associated with impaired olfactory capacities (Paß et al., 2020). However, neither mitochondrial density nor metabolic capacity influenced prey detection in *L. delicata*. In contrast, our results suggests that metabolic function in the brain is not a limiting factor for prey detection in *L. delicata*, or that *L. delicata* can sustain this cognitive function despite fluctuations in energy availability.

Similarly, oxidative stress and oxidative damage did not influence the ability of lizards to detect visual or chemical stimuli from prey. Oxidative stress and its associated damage can decrease cognitive abilities through neuron death or senescence (Du et al., 2009; Finkel & Holbrook, 2000; Gong et al., 2011; Hoffmann & Spengler, 2018; Zhu et al., 2004). However, we found no relationship between ROS, lipid peroxidation, or DNA damage and prey detection abilities. Possibly, the levels of oxidative damage observed in our study were not sufficient to impair prey detection either because of compensatory mechanisms—such as antioxidant activity—that counteracted the negative effects of oxidative stress, or because the brain regions we studied had a high neural density that provided functional resilience (Amiel et al., 2011; Rice et al., 2002). Alternatively, the cognitive effects of oxidative stress may not be detectable at the age when we tested lizards. Oxidative damage accumulates over time, and its effects on cognition become more pronounced with age (Hara et al., 2014; Terman & Brunk, 2006). As such, we could find strong relationships between oxidative damage and perception if older individuals were tested. In fact, we found a positive relationship between age and DNA damage in olfactory bulbs, and between age and lipid peroxidation in both brain regions. Cross-sectional studies across life stages may help reveal how oxidative damage shapes cognitive function over time.

Conclusions

Prenatal CORT exposure and incubation temperature produced region-specific effects in the brain of *L. delicata*. While mitochondrial density, metabolic capacity, and ROS production remained remarkably resilient to prenatal conditions, oxidative damage and behavioural responses varied depending on treatment and brain region. The interaction between prenatal CORT exposure and incubation temperature influenced DNA damage and lipid peroxidation in the olfactory bulbs and the optic tecta. However, these effects were not related to the perception of visual or chemical prey stimuli. Notably, chemical perception was affected by prenatal CORT alone, highlighting a selective sensitivity of perceptual abilities independent of measured brain physiology. Our findings suggest that differences in prey detection may result from treatment effects on other factors, like motivation. Future studies should explore how the environment influences mitochondrial function accross tissues, as well as the mechanisms underlying the brain's resilience to early-life conditions.

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